

Lab 1 – Use-Case Flow Specification

Problem Statement #28: Inter-City Freight Load Matching Marketplace

Use Case: Submit Bid for Posted Freight Shipment

- Primary Actor: Freight Carrier
- Supporting Actor: Shipper (receives bid notification)

Preconditions

- A freight shipment has been posted by a shipper and is within its specified auction timeframe.
- The freight carrier is verified and has valid permits.

Postconditions

- The valid carrier bid is recorded against the posted shipment.
- The shipper is notified of the submitted bid.

Main Success Scenario

1. Carrier selects an available posted freight shipment.
2. System displays the shipment details and the remaining auction timeframe.
3. Carrier enters the proposed bid details.
4. System verifies that the carrier is verified and has valid permits.
5. System validates the bid and confirms that it is submitted within the auction timeframe.
6. System records the carrier's bid for the shipment.
7. System notifies the shipper that a new bid has been submitted.
8. Use case ends successfully.

Alternate Flow – Unverified Carrier / Invalid Permits

- 4a. System finds that the carrier is not verified or does not have valid permits.
- 4a1. System rejects the bid submission and displays a message explaining that only verified carriers with valid permits may bid.
- 4a2. No bid is recorded and the shipper is not notified.
- 4a3. Use case ends unsuccessfully.